

# HiGrEC: A Controlled Study of Structural Inductive Bias for Response Clarity Classification in Political Interviews

Vidvathama R   Siddarth Gottumukkula   Sanjana Reddy Vonteri   Shashikanta Sahoo  
2024122002   2023102040   2026901007   2026900007

Team Nier\_ANLP  
International Institute of Information Technology, Hyderabad

## Abstract

SemEval-2026 Task 6 asked systems to classify political interview answers by clarity and by evasion technique. The coarse subtask was pushed well past its baseline; the fine-grained nine-way subtask was not moved at all, and encoder-only systems saturated regardless of scale. We argue this reflects two measurement failures rather than limited capacity. First, the coarse taxonomy is cut across the wrong axis: CLEAR REPLY holds one leaf while AMBIVALENT holds five spanning full, partial and zero information delivery and disagreement concentrates where that cut crosses the delivery dimension. Second, the scorer is multi-reference with an asymmetric penalty, so the optimal prediction is the label most likely to lie *inside* the annotator set rather than the likeliest consensus label, which no system pursues. We propose an attribute code over the nine leaves, validated by re-annotation, and a decision rule estimating set-membership probability through a learned annotator model.

## 1 Project Idea

Response clarity classification asks whether a politician’s answer delivered what the question requested. Given an interview question and answer, a system predicts a coarse clarity label (CLEAR REPLY, AMBIVALENT, CLEAR NON-REPLY) and one of nine nested evasion techniques (Thomas et al., 2024); SemEval-2026 Task 6 made this a shared task (Thomas et al., 2026).

Our project proposes that the fine-grained subtask has stalled for reasons outside the model: the label space discards the structure the taxonomy encodes, and the metric rewards a quantity nobody optimises. We aim to (i) show empirically that the coarse taxonomy is cut along the wrong axis, (ii) replace the flat output space with an attribute code recovering that axis, and (iii) derive the decision rule the official scorer implies. The unifying claim

is that this task is limited by how it was set up, and not by how hard the models try.

## 2 Motivation: Shortcomings of Existing Work

Roughly a hundred teams attacked the same data with the full contemporary toolkit prompted frontier models, staged reasoning, confidence-gated verification, encoder ensembles, synthetic augmentation and the results are lopsided. The coarse subtask advanced well past the organisers’ baseline; the fine-grained subtask did not. The winning system merely matched the strongest baseline, the median entry sat far below it, and a fifteen-team band was separated by less than seed noise. Encoder-only systems plateaued regardless of scale.

When a hundred independent attempts using different methods converge on one ceiling, the limitation is unlikely to be any one method. We identify four shortcomings shared across the published systems.

**(G1) The output space is flat.** Every system treats the nine leaves as nine unrelated categories, either as a flat softmax or as nine names in a prompt. Nothing lets a leaf with fewer than a hundred training examples borrow strength from a frequent leaf it shares structure with. The organisers additionally report that loss-side remedies for imbalance - class weighting, focal loss underperform plain cross-entropy here, which closes the obvious alternative and leaves the output space as the remaining lever.

**(G2) The taxonomy is never questioned.** Every system takes the label set as given and asks how to predict it better. The winning system’s errors are not diffuse: over three-quarters fall in a single three-class neighbourhood, and one class has recall below 0.2 despite high precision - the model cannot find it at all. The dataset creators report that the

same neighbourhood is where annotators disagree most, and that some of that disagreement crosses the coarse boundary. A taxonomy whose errors and whose disagreements coincide this precisely is worth interrogating rather than fitting.

**(G3) Annotator identity and disagreement are discarded.** The training split records which annotator produced each label, and the evaluation splits carry multiple independent annotations. Every surveyed system collapses this to a single gold label and treats residual disagreement as noise. This is a substantial loss: the disagreement is concentrated in exactly the classes that fail, and substantial work argues such variation is signal rather than error (Plank, 2022; Uma et al., 2021).

**(G4) Systems optimise a different quantity than the scorer rewards.** The fine-grained metric is multi-reference: a prediction counts as correct if it matches any annotator’s label, and - crucially - one landing inside the annotator set incurs no penalty for the set members it did not predict, while one outside is penalised once for every label in the set. The cost of an error therefore scales with how much the annotators disagreed. Under such a metric the Bayes-optimal prediction is the label most likely to appear *somewhere* in the set, which is not the most likely consensus label. Every published system predicts the latter.

**Contributions.** Each contribution is an attempt on one of these gaps.

**C1 addresses G1.** We replace the flat output space with a code over five attributes coverage, engagement, form, marker and incompleteness in which each leaf is a unique combination but most attribute values are shared across several leaves. A rare leaf then discriminates on top of a representation estimated from thousands of examples rather than its own handful: a structural response to imbalance that never touches the loss, which matters given that loss-side remedies fail here.

**C2 addresses G2.** We argue, then test, that the coarse taxonomy is cut across the coverage axis and that this predicts the observed agreement pattern. The test is a re-annotation study measuring whether agreement rises under a coverage-derived coarse cut. A negative result is as informative as a positive one, and we commit to reporting either.

**C3 addresses G3.** We model the observed label as a latent label passed through an annotator-specific confusion structure, recovering a bias-free

posterior and characterise what is *not* recoverable, since full per-annotator matrices are unidentifiable from singly-labelled data. Reporting that limit is part of the contribution.

**C4 addresses G4.** We derive the rule the scorer implies: estimate the probability each class lies in the annotator set, by pushing the latent posterior through C3’s annotator model, and select under per-class thresholds tuned for macro-F1. C3 therefore stops being discarded at inference and becomes what makes C4 computable.

**C5 addresses G2.** We compute an explicit coverage-and-engagement representation of the answer relative to the question, since the failing neighbourhood is a disagreement along those two dimensions.

### 3 Background

Political theory treats response ambiguity as strategy rather than failure. Bavelas et al. (1988) argue that equivocation emerges when every direct answer carries a cost, and Bull (2003) finds politicians reply clearly to a minority of interview questions, far below the rate for non-politicians. Clayman (2001) shows the boundary between responsiveness and evasion is shaped by institutional norms rather than by any simple linguistic marker.

What matters here is that this literature holds evasion to be *structured*: speakers draw on a recurring, enumerable repertoire, which is what makes a taxonomy possible at all. The classical typologies - notably Bull and Mayer (1993) organise that repertoire along an information-delivery axis: full replies, intermediate replies, and non-replies subdivided by technique.

Thomas et al. (2024) bring this into NLP, consolidating those typologies into a two-level taxonomy and releasing QEvasion. Two design choices are decisive here. They replace the reply/non-reply axis with a clarity/ambiguity axis, on the grounds that judging clarity is less bias-prone for non-expert annotators than judging validity; and they eliminate the intermediate-reply category by decomposing multi-part questions. Both are reasonable and well argued. Together, however, they remove the coverage dimension from the label space and it is that removal we propose to examine.

### 4 Related Work

**Evasion detection.** Thomas et al. (2024) establish the task; Thomas et al. (2026) report the

shared task. The strongest system (Shi et al., 2026) routes examples through confidence-gated reasoning stages with confusion-mined demonstrations, deriving the coarse label from the leaf. The competition’s clearest finding is that *any* hierarchical decomposition beats flat inference 0 indirect evidence for our position, since systems benefit from taxonomy structure while re-deriving it every time.

**Answer intent and answerability.** Ferracane et al. (2021) is the nearest prior work: congressional testimony with a two-level act-to-intent hierarchy and explicit modelling of annotator subjectivity. We differ in target 0 they model perceived intent, we model delivered information 0 but any structural claim we make must be differentiated from theirs. Louis et al. (2020) on indirect answers supplies the closest analogue to our coverage dimension and a candidate transfer source.

**Learning under disagreement.** Our annotator model descends from Dawid and Skene (1979) through the crowd layer of Rodrigues and Pereira (2018), regularised after Tanno et al. (2019), within the perspectivist line of Plank (2022) and Uma et al. (2021); see also Davani et al. (2022). We claim no novelty for the mechanism.

**Output spaces and decision theory.** The attribute code is standard attribute-based classification (Palatucci et al., 2009) transplanted into a discourse taxonomy. The decision rule builds on plug-in F-measure maximisation (Lipton et al., 2014; Koyejo et al., 2014) and must be distinguished from set-valued *prediction* (Mortier et al., 2021), the opposite orientation: there the system outputs a set, here the *gold* is one. Each ingredient of C4 is standard; the composition is what we believe new, and our only novelty claim.

## 5 Proposed Methodology

**Attribute code (C1).** Each leaf definition is a conjunction of a few judgments. *Coverage* records how much requested information arrived (full, partial, none); *engagement* whether the answer stays on target, pivots, or ignores the question; *form* distinguishes stated from implied delivery; *marker* records an overt signal of non-delivery; *incompleteness* distinguishes missing components from missing specificity. Every leaf receives a unique code, while coverage, engagement and form take values shared by several leaves and supported by large

fractions of the data. The leaf posterior is the product of attribute posteriors restricted to the nine valid codes, and the coarse label is derived from the leaf.

We are explicit about the limit: nine leaves need nine distinct codes, so each must be uniquely identified by *some* value, and for rare leaves that value is itself rare. The code does not eliminate imbalance; it changes what fraction of a rare leaf’s representation is well estimated.

**Annotator model (C3).** We treat the observed label as a latent label passed through an annotator-specific confusion structure and maximise the likelihood of observed labels. Because training items were distributed at random, differences between annotators’ marginals are attributable to annotator effects rather than item effects, identifying a constrained model 0 a marginal-bias vector, or a low-rank confusion with a prior pulling toward the identity. The multiply-annotated split permits direct estimation of a global confusion matrix. A full matrix per annotator is *not* identifiable from singly-labelled data, and we report that rather than obscure it.

**Decision rule (C4).** Let  $q(c | x)$  be the probability that class  $c$  appears in the annotator set, obtained by pushing the latent posterior through the annotator model; it is not a distribution, since expected set size exceeds one. Because macro-F1 is not decomposable across examples, we select with a plug-in rule under per-class thresholds fitted on held-out data,  $\hat{y} = \arg \max_c \lambda_c q(c | x)$ , the thresholds absorbing the unequal per-class F1 gradients created by the long tail.

**Coverage estimation (C5).** Because each sub-question is paired with the full original answer, the problem is locating which span, if any, responds to it 0 span selection with a NONE option. The strongest published system performs exactly this check in prose, as a key-information override in its prompt; we make it a trained module instead, supervised by distilled span judgments, and read coverage and engagement statistics off the result. This targets the failing neighbourhood directly.

The two rules diverge exactly when a class is often *in* the annotator set but rarely its mode. Those are the highest-disagreement classes, which are also where the field is failing, so the method predicts where its own gains must appear.

**Backbones.** Since encoders are reported saturated here, parameter count is not the contribution.

The primary system is a LoRA-tuned open-weight LLM of moderate size, so gains are attributable to the mechanism rather than scale, with a base-size encoder as an efficiency reference. Inference is single-pass, making cost comparisons against multi-call pipelines meaningful.

## 6 Dataset

QEvasion (Thomas et al., 2024) comprises question–answer pairs from several hundred US presidential interviews spanning nearly two decades and four administrations. Multi-barrelled questions were decomposed into single sub-questions using an LLM, and three non-expert annotators labelled each sub-response under a political-science expert acting as validator and tie-breaker.

Three properties drive our design. Training items carry a single raw annotator judgment *together with the identity of the annotator who produced it*, which makes C3 possible and which no prior system uses. The evaluation splits carry multiple independent annotations per item, which makes C4’s metric meaningful and lets the learned confusion be validated against a measured one. And the leaf distribution is long-tailed, which is what makes C1 necessary.

**Re-annotation.** Testing C2 requires data the release does not contain: independent judgments of the attributes themselves. We will re-annotate several hundred items with four to five annotators labelling every item, collecting the leaf label, the attribute code assigned independently, and annotator identity, reusing the original protocol including its counterfactual-injection check. One campaign serves three purposes 0 validating C1, supplying multiply-annotated items for C3 and C4, and releasing a resource 0 and must be designed for all three from the outset.

## 7 Evaluation Plan

**Baselines.** A flat nine-way softmax over the same backbone, identical in every other respect and given a matched hyperparameter budget, is the reference every component is measured against; we freeze that checkpoint and gate against it. We add a parameter-matched capacity control and 0 for C4 0 majority-vote gold with standard thresholding, separating the decision-theoretic gain from the modelling gain. Published systems are context, not like-for-like comparisons.

**Protocol.** Five or more seeds with mean and standard deviation on every number; bootstrap intervals and per-class support alongside every per-class F1; cross-validation on the training split as primary evaluation, since the development split leaves the rarest classes with single-digit support; and paired bootstrap significance. The distribution shift is handled by importance weighting.

**Ablations and analysis.** The heavy-weight contrasts are the code versus flat on rare leaves; a shuffle control destroying question–answer correspondence in the span module, separating structure from capacity; and the decision rule applied to a *fixed* model. We also report the confusion sub-matrix over the failing neighbourhood before and after C5, coverage- and engagement-factor accuracy separately (a decomposition no prior system can report), and accuracy stratified by annotator consensus, revealing whether gains arrive on ambiguous items or only easy ones.

**Advance predictions and success.** To make the ablations falsifiable we commit in advance. C1 should help the rare leaf whose distinguishing attributes are otherwise well populated, and *not* the three leaves with overt lexical markers that published systems already classify near-perfectly; helping those instead would mean it adds capacity rather than shares structure. C4’s gains should fall on non-unanimous items and the two highest-disagreement classes; gains on unanimous items would falsify the account. We state no macro-F1 threshold, since single-digit support for the rarest leaves puts any small margin below the measurement resolution. The project succeeds if the decision rule beats consensus selection with the gain where predicted, the re-annotation answers definitively whether a coverage-derived cut raises agreement, and the shuffle control shows structure drives C5.

## References

- Janet Beavin Bavelas, Alex Black, Lisa Bryson, and Jennifer Mullett. 1988. Political equivocation: A situational explanation. *Journal of Language and Social Psychology*, 7:137–145.
- Peter Bull. 2003. *The Microanalysis of Political Communication: Claptrap and Ambiguity*. Routledge.
- Peter Bull and Kate Mayer. 1993. How not to answer questions in political interviews. *Political Psychology*, 14:651–666.
- Steven E. Clayman. 2001. Answers and evasions. *Language in Society*, 30(3):403–442.

- Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the ACL*, 10:92–110.
- A. Philip Dawid and Allan M. Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society C*, 28(1):20–28.
- Elisa Ferracane, Greg Durrett, Junyi Jessy Li, and Katrin Erk. 2021. Did they answer? subjective acts and intents in conversational discourse. In *NAACL-HLT*, pages 1626–1644.
- Oluwasanmi Koyejo, Nagarajan Natarajan, Pradeep Ravikumar, and Inderjit S. Dhillon. 2014. Consistent binary classification with generalized performance metrics. In *NeurIPS*.
- Zachary C. Lipton, Charles Elkan, and Balakrishnan Naryanaswamy. 2014. Optimal thresholding of classifiers to maximize F1 measure. In *ECML-PKDD*, pages 225–239.
- Annie Louis, Dan Roth, and Filip Radlinski. 2020. “I’d rather just go to bed”: Understanding indirect answers. In *EMNLP*, pages 7411–7425.
- Thomas Mortier, Marek Wydmuch, Krzysztof Dembczyński, Eyke Hüllermeier, and Willem Waegeman. 2021. Efficient set-valued prediction in multi-class classification. *Data Mining and Knowledge Discovery*, 35:1435–1469.
- Mark Palatucci, Dean Pomerleau, Geoffrey Hinton, and Tom Mitchell. 2009. Zero-shot learning with semantic output codes. In *NeurIPS*.
- Barbara Plank. 2022. The “problem” of human label variation: On ground truth in data, modeling and evaluation. In *EMNLP*, pages 10671–10682.
- Filipe Rodrigues and Francisco Pereira. 2018. Deep learning from crowds. In *AAAI*, pages 1611–1618.
- Lingling Shi, Haoyu Jin, Shiquan Wang, Ruiyu Fang, Shuangyong Song, and Xuelong Li. 2026. TeleAI at SemEval-2026 task 6: A confidence-aware multi-stage reasoning framework with chain-of-thought. In *Proceedings of SemEval-2026*, pages 1591–1609.
- Ryutaro Tanno, Ardavan Saeedi, Swami Sankaranarayanan, Daniel C. Alexander, and Nathan Silberman. 2019. Learning from noisy labels by regularized estimation of annotator confusion. In *CVPR*.
- Konstantinos Thomas, Giorgos Filandrianos, Maria Lymperaiou, Chrysoula Zerva, and Giorgos Stamou. 2024. “I never said that”: A dataset, taxonomy and baselines on response clarity classification. In *Findings of EMNLP*, pages 5204–5233.
- Konstantinos Thomas, Giorgos Filandrianos, Maria Lymperaiou, Chrysoula Zerva, and Giorgos Stamou. 2026. SemEval-2026 task 6: CLARITY – unmasking political question evasions. In *Proceedings of SemEval-2026*.
- Alexandra Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470.